Machine learning in agriculture

Introduction

- **Smart farming** powered by ML
- Tackles **food security, climate change, resource optimization**
- Data-driven decisions from sensors, drones, and satellites

MI process

ML Process in Agriculture

- **Data collection** – soil sensors, drones, satellites
- **Data cleansing** – remove errors for reliability
- **Feature engineering** – extract soil pH, rainfall, crop type
- **Model selection** – regression, decision trees, neural networks
- **Testing & validation** – check accuracy
- **Deployment** – apply in real farms, update continuously

Impacts in agriculture

Impacts

- **Higher productivity** – optimized irrigation/fertilization
- **Cost reduction** – less waste of resources
- **Sustainability** – eco-friendly farming practices
- **Food security** – stable yields for global demand
- **Market efficiency** – better price predictions

challenges

Data availability – small farmers lack access

High costs – advanced ML systems are expensive

Technical expertise – need training

Infrastructure gaps – poor connectivity in rural areas

Model accuracy – depends on quality data

Future updates

Future Updates

- **Smart irrigation** – AI-driven water management
- **Climate resilience** – adaptive farming strategies
- **Robotics** – automated harvesting/planting
- **Blockchain + ML** – transparent supply chains
- **Genomics + ML** – climate-resistant crop breeding

conclusion

ML makes agriculture **smarter, efficient, sustainable**

Challenges remain: cost, infrastructure, expertise

Future promises **AI-powered farms** feeding the world sustainably